

# Technical Approach: Multi-Layer AI Block Planning Platform

Smart India Hackathon • Ministry of Railways (PS 26027) • Corridor: New Delhi (NDLS) → Kanpur Central (CNB)

G&SR 2026 STATUTORY COMPLIANT PROVEN IN SILICO

## 1. INGESTION & AI PRIORITIZATION (L1 & L2)

- **Live Telemetry:** Sub-minute train positions & speeds via **CRIS COA API** and **ISRO RTIS GPS**.
- **Tri-Dept Requisitions:** Aggregates demands from **TMS** (P-Way track defects), **SMMS** (points/signals), and **TDMS** (25kV OHE catenary).
- **TG-4 Track Car CSVs:** Automatic parsing of geometry inspection runs into normalized Track Quality Indices (TGI).
- **ML Risk Scoring:** **XGBoost 2.0** ranks defects into Priority Scores (0–100) using GMT tonnage, track age, and speed limits.
- **Explainable AI:** **SHAP** feature attribution + **Rail Sarthi Copilot (Gemini)** with <5ms read-only DFA guardrails.

## 2. DISCRETE CP-SAT OPTIMIZER (L3)

- **Constraint Solver:** Powered by **Google OR-Tools 9.8 CP-SAT** solving exact non-linear scheduling across 24h horizons.
- **Headway Lull Detection:** Discovers off-peak traffic gaps between high-speed express trains (Vande Bharat, Rajdhani).
- **Multi-Dept Bundling:** Mathematically merges P-Way, S&T, and OHE requests in adjacent KMs into single **Super-Blocks**.
- **Heavy Machine Chaining:** Routes track tampers (CSM), ballast cleaners (BCM), and tower wagons without yard dead-runs.
- **Dynamic What-If Re-planner:** Sub-8.5s delta re-solve upon emergencies (rail fractures, train delay +45m) with frozen zones.

## 3. STATUTORY SAFETY & COCKPITS (L4 & L5)

- **Disconnection Memo (Form T/351):** Enforces statutory Station Master route lockouts and virtual switch collars before work.
- **Power Permit (TPC PTW):** Coordinated 25kV catenary power block de-energization handshake with electrical teams.
- **Kavach TCAS L2 Envelope:** Transmits digital red possession signals directly to cab radios, preventing train approach.
- **Security Vault:** PostgreSQL 15 + PostGIS spatial topology + Redis 7.0 refresh token rotation & revocation blacklist.
- **PKI Audit & 4-Tier UI:** Ed25519 cryptographic CRS judicial audit trail. **React 18 + Vite + MUI** cockpits with COA String Charts.

## END-TO-END OPERATIONAL LIFECYCLE FLOW (FROM DEFECT DETECTION TO TRACK RECONNECTION)

